Beyond First-Order Methods for Distributed Optimization

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May 5, 2023

Outline

1. Problem Setup
2. Hardness for Second-Order Distributed Optimization
3. C2EDEN: Communication and Computation Efficient Distributed Newton Method
4. Future Work

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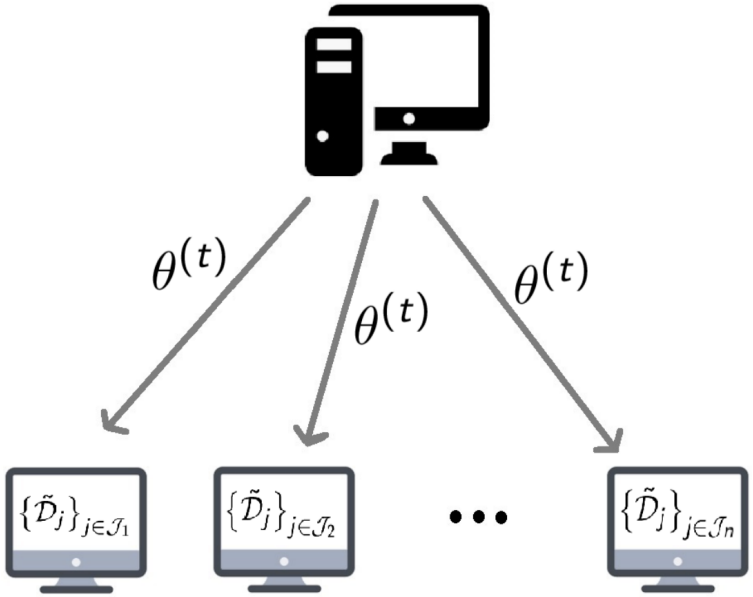
Settings

We consider the **centralized** distributed optimization.

$$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} f_i(\mathbf{x}), \tag{1}$$

where n is the numbers of the devices, each f_i is twice-differentiable.



Single agent optimization

We first consider single-agent optimization of the form

$$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x}). \tag{2}$$

We define the oracle

$$T_{p,L_p}(\mathbf{x}) := \arg\min_{\mathbf{y}} \underbrace{f(\mathbf{x}) + \sum_{i=1}^{p} \frac{1}{i!}\nabla^i f(\mathbf{x})[\mathbf{y}-\mathbf{x}]^i}_{f_p(\mathbf{y};\mathbf{x})\text{:Taylor expansion at } \mathbf{x}} + \frac{L_p}{(p+1)!}\|\mathbf{y}-\mathbf{x}\|^{p+1}.$$

Proposition

If $\nabla^p f(\mathbf{x})$ is L_p-Lipschitz continuous, i.e.,

$$\|\nabla^p f(\mathbf{x}) - \nabla^p f(\mathbf{y})\| \le L_p\|\mathbf{x}-\mathbf{y}\|,$$

then we always have $f(\mathbf{y}) \le f_p(\mathbf{y};\mathbf{x})$.

First/Second-order methods for single agent optimization

$$T_{p,L_p}(\mathbf{x}) := \arg\min_{\mathbf{y}} f(\mathbf{x}) + \sum_{i=1}^{p} \frac{1}{i!}\nabla^i f(\mathbf{x})[\mathbf{y}-\mathbf{x}]^i + \frac{L_p}{(p+1)!}\|\mathbf{y}-\mathbf{x}\|^{p+1}.$$

- First-order method:

$$\mathbf{x}_{t+1} = T_{1,L_1}(\mathbf{x}_t) = \mathbf{x}_t - \frac{1}{L_1}\nabla f(\mathbf{x}_t)$$

- Second-order method:

$$\mathbf{x}_{t+1} = T_{2,L_2}(\mathbf{x}_t) = \mathbf{x}_t - \left(\nabla^2 f(\mathbf{x}_t) + L_2 r_t \mathbf{I}_d\right)^{-1} \nabla f(\mathbf{x}_t)$$

  for the simplicity, we write Newton method in the following form in the left talk

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \left(\nabla^2 f(\mathbf{x}_t)\right)^{-1} \nabla f(\mathbf{x}_t)$$

First/Second-order methods for single agent optimization

Second-order methods enjoy a superior convergence rates than the first-order methods!

Convergence behavior of first and second-order methods

Methods	Nonconvex	Strongly-convex
Gradient descent	$\mathcal{O}(\epsilon^{-2})$	linear
(Cubic-regularized) Newton	$\mathcal{O}(\epsilon^{-3/2})$	superlinear

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First-order methods for distributed optimization

Let's return to the centralized distributed optimization.

$$\min_{\mathbf{x}\in\mathbb{R}^d} f(\mathbf{x}) = \frac{1}{n}\sum_{i=1}^{n} f_i(\mathbf{x}) \tag{3}$$

Distributed gradient descent is very popular for many tasks.

- The client i: calculate $\nabla f_i(\mathbf{x}_t)$
- The server: $\mathbf{x}_{t+1} = \mathbf{x}_t - \frac{1}{L_1}\underbrace{\left(\frac{1}{n}\sum_{i=1}^{n}\nabla f_i(\mathbf{x}_t)\right)}_{\text{received from each client}} = \mathbf{x}_t - \frac{1}{L_1}\nabla f(\mathbf{x}_t)$

Communication cost at each iteration: $\mathcal{O}(d)$ $(\nabla f_i(\mathbf{x}_t), \mathbf{x}_{t+1})$

Hardness for second-order for distributed optimization

Consider distributed (cubic-regularized) Newton method.

- The client i: calculate $\nabla f_i(\mathbf{x}_t)$, $\nabla^2 f_i(\mathbf{x}_t)$
- The server:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \left(\underbrace{\frac{1}{n}\sum_{i=1}^{n} \nabla^2 f_i(\mathbf{x}_t)}_{\text{received from each client}} \right)^{-1} \left(\underbrace{\frac{1}{n}\sum_{i=1}^{n} \nabla f_i(\mathbf{x}_t)}_{\text{received from each client}} \right)$$
$$= \mathbf{x}_t - [\nabla^2 f(\mathbf{x}_t)]^{-1} \nabla f(\mathbf{x}_t)$$

Communication cost at each iteration $\mathcal{O}(d^2)$ ($\nabla^2 f_i(\mathbf{x}_t)$, $\nabla f_i(x_t)$ and $\mathbf{x}_{t+1}$)
Communicating the Hessian is bad for distributed optimization!

Designing communication efficient second-order methods

Communication Efficient: At each iteration, the communication cost between the server and the client is $\mathcal{O}(d)$.

Local Descent (GIANT/Local-Cubic):

- The client i: $\mathbf{x}_{i,t+1} = \mathbf{x}_t - \nabla^2 f_i(\mathbf{x}_t)^{-1} \nabla f(\mathbf{x}_t)$
- The server: $\mathbf{x}_{t+1} = \underbrace{\frac{1}{n}\sum_{i=1}^{n} \mathbf{x}_{i,t+1}}_{\text{received from each client}} \neq \mathbf{x}_t - [\nabla^2 f(\mathbf{x}_t)]^{-1} \nabla f(\mathbf{x}_t)$

These methods do not enjoy the superiorty of second-order methods (neither $\mathcal{O}(\epsilon^{-3/2})$ for non-convex nor superlinear for strongly-convex).

Designing communication efficient second-order methods

Solve quadratic sub-problems: The (Cubic)-Newton step can be reformulated to

$$\min_{\mathbf{y}} f(\mathbf{x}_t) + \langle \nabla f(\mathbf{x}_t), \mathbf{y} - \mathbf{x}_t \rangle + \frac{1}{2}(\mathbf{y} - \mathbf{x}_t)^\top \nabla^2 f(\mathbf{x}_t)(\mathbf{y} - \mathbf{x}_t), \quad (4)$$

The gradient of equation (4) at $\mathbf{y}$ is

$$\nabla f(\mathbf{x}_t) + \nabla^2 f(\mathbf{x}_t)(\mathbf{y} - \mathbf{x}_t) = \frac{1}{n}\sum_{i=1}^{n} \nabla f_i(\mathbf{x}_t) + \left(\frac{1}{n}\sum_{i=1}^{n} \nabla^2 f_i(\mathbf{x}_t)\right)(\mathbf{y} - \mathbf{x}_t)$$

which can be communicated in $\mathcal{O}(d)$.

Solving the quadratic sub-problem at each iteration will increase the total cost and leads to complicated implementation.

Designing communication efficient second-order methods

Compressed Newton methods: Doing compression before sending the second-order information to the server. These methods enjoy local superlinear rate, however

- it requires constructing local Hessian estimator at each client which raise the space complexity to $\mathcal{O}(d^2)$
- No theoretical results for non-convex case.

Designing communication efficient second-order methods

Method	Communication	Space	NC	SC
Distributed-GD	$\mathcal{O}(d)$	$\mathcal{O}(d)$	$\mathcal{O}(\epsilon^{-2})$	linear
Distributed-Newton	$\mathcal{O}(d^2)$	$\mathcal{O}(d^2)$	$\mathcal{O}(\epsilon^{-3/2})$	superlinear
GIANT/DANE	$\mathcal{O}(d)$	$\mathcal{O}(d)$	$\mathcal{O}(\epsilon^{-2})$	linear
Distributed sub Newton	$\mathcal{O}(d)$	$\mathcal{O}(d)$	unknown	linear
Compressed Newton	$\mathcal{O}(d)$	$\mathcal{O}(d^2)$	unknown	superlinear

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C2EDEN

We developed Communication and Computation Efficient Distributed Newton Method which fixed the previous gaps.

Method	Communication	Space	NC	SC
Distributed-GD	$\mathcal{O}(d)$	$\mathcal{O}(d)$	$\mathcal{O}(\epsilon^{-2})$	linear
Distributed-Newton	$\mathcal{O}(d^2)$	$\mathcal{O}(d^2)$	$\mathcal{O}(\epsilon^{-3/2})$	superlinear
GIANT/DANE	$\mathcal{O}(d)$	$\mathcal{O}(d)$	$\mathcal{O}(\epsilon^{-2})$	linear
Distributed sub Newton	$\mathcal{O}(d)$	$\mathcal{O}(d)$	unknown	linear
Compressed Newton	$\mathcal{O}(d)$	$\mathcal{O}(d^2)$	unknown	superlinear
C2EDEN	$\mathcal{O}(d)$	$\mathcal{O}(d)$	$\mathcal{O}(\epsilon^{-3/2})$	**superlinear**

Observations

- Communicate a full Hessian is difficult, but we can atomize the Hessian

$$\nabla^2 f_i(\tilde{\mathbf{x}}) = \begin{bmatrix} | & | & & | \\ \nabla^2 f_i(\tilde{\mathbf{x}})\mathbf{e}_1 & \nabla^2 f_i(\tilde{\mathbf{x}})\mathbf{e}_2 & \cdots & \nabla^2 f_i(\tilde{\mathbf{x}})\mathbf{e}_d \\ | & | & & | \end{bmatrix}$$

  where communicating $\nabla^2 f_i(\tilde{\mathbf{x}})\mathbf{e}_p \in \mathbb{R}^d$ (a Hessian-vector) requires only $\mathcal{O}(d)$.
- We can communicate a full Hessian in d iterations ($\frac{d^2}{d} = d$)
- We can communicate the full Hessian in d iterations while updating the model!

C2EDEN: algorithm

At iteration $k = td, td+1 \cdots, (t+1)d-1$,

- The client: compute $\nabla f_i(\mathbf{x}_k)$, $\nabla^2 f_i(\mathbf{x}_{td})\mathbf{e}_{k-td+1}$
- The server:

$$\mathbf{H}(:, k-td+1) = \underbrace{\frac{1}{n}\sum_{i=1}^{n} \nabla^2 f_i(\mathbf{x}_{td})\mathbf{e}_{k-td+1}}_{\text{from current iteration}}$$

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \left[\underbrace{\nabla^2 f(\mathbf{x}_{(t-1)d})}_{\text{from iteration } (t-1)d \text{ to } td-1}\right]^{-1} \left(\underbrace{\frac{1}{n}\sum_{i=1}^{n} \nabla f_i(\mathbf{x}_k)}_{\text{from current iteration}}\right)$$

Communication cost at each iteration: $\mathcal{O}(d)$ $(\nabla^2 f_i(\tilde{\mathbf{x}})\mathbf{e}_p,\ \nabla f_i(\mathbf{x}_t),\ \mathbf{x}_{t+1})$

C2EDEN: comparison

At iteration $k = td, td+1 \cdots, (t+1)d-1$,

C2EDEN

- The client: compute $\nabla f_i(\mathbf{x}_k)$, $\nabla^2 f_i(\mathbf{x}_{td})\mathbf{e}_{k-td+1}$
- The server:

$$\mathbf{H}(:, k-td+1) = \frac{1}{n}\sum_{i=1}^{n} \nabla^2 f_i(\mathbf{x}_{td})\mathbf{e}_{k-td+1}$$

$$\mathbf{x}_{t+1} = \mathbf{x}_t - [\nabla^2 f(\mathbf{x}_{(t-1)d})]^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\nabla f_i(\mathbf{x}_k)\right)$$

Distributed-Newton

- The client: compute $\nabla f_i(\mathbf{x}_k)$, $\nabla^2 f_i(\mathbf{x}_k)$
- The server:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - [\nabla^2 f(\mathbf{x}_k]^{-1}\left(\frac{1}{n}\sum_{i=1}^{n}\nabla f_i(\mathbf{x}_k)\right)$$

C2EDEN: convergence results

Theorem (Nonconvex case)

Suppose the Hessian of the object function f is L_2-Lipschitz continuous, then with no more than $\mathcal{O}(\sqrt{dL_2}\epsilon^{-3/2})$, C2EDEN finds a stationary point $\mathbf{x}_i$ satisfied

$$\|\nabla f(\mathbf{x}_i)\| \leq \epsilon \quad \textit{and} \quad \lambda_{\min}(\nabla^2 f(\mathbf{x}_i)) \geq -\sqrt{dL_2\epsilon}$$

Theorem (Strongly-convex case)

Suppose the Hessian of the object function f is L_2-Lipschitz continuous and f is μ-strongly convex, then C2EDEN converges at a superlinear rates locally, i.e.,

$$\lim_{k\to\infty} \frac{\|\nabla f(\mathbf{x}_{k+1})\|}{\|\nabla f(\mathbf{x}_k)\|} = 0$$

C2EDEN: empirical study

We validate C2EDEN on regularized logistic regression.

$$f_i(\mathbf{x}) = \frac{1}{m_i} \sum_{j=1}^{m_i} \ln \left(1 + \exp\left(-b_{ij} \mathbf{x}^\top \mathbf{a}_{ij}\right)\right) + \lambda R(\mathbf{x}), \tag{5}$$

For the nonconvex case, we set

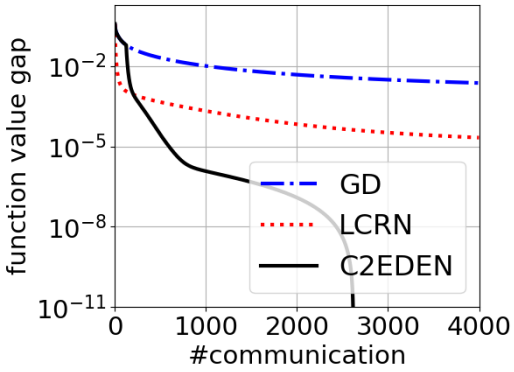
$$R(\mathbf{x}) = \sum_{p=1}^{d} \frac{x_{(p)}^2}{1 + x_{(p)}^2}$$

For the strongly convex case, we set

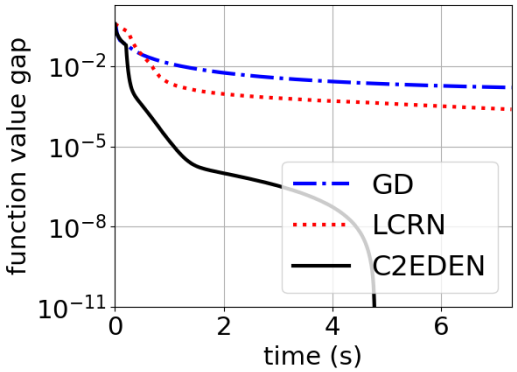
$$R(\mathbf{x}) = \|\mathbf{x}\|_2^2$$

C2EDEN: empirical study

Nonconvex case:



(a) #communication vs. gap

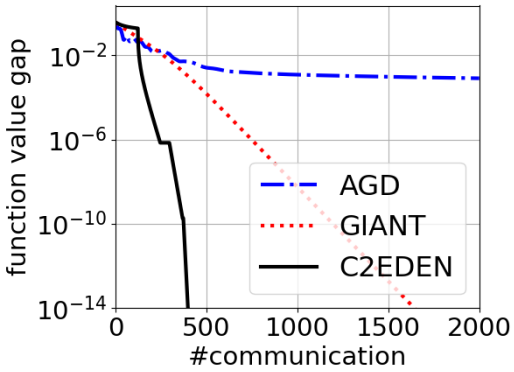


(b) time (s) vs. gap

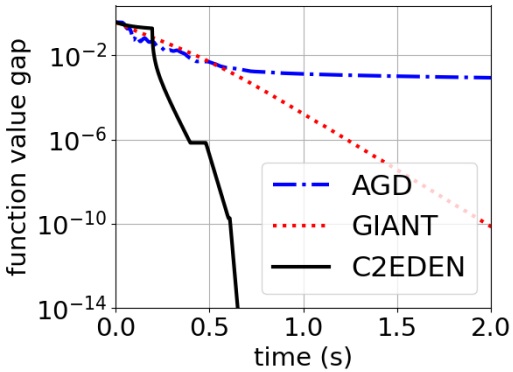
The results of the model of nonconvex regularized logistic regression on "a9a".

C2EDEN: empirical study

Strongly-convex case:



(a) #communication vs. gap



(b) time (s) vs. gap

The results of the model of l_2-regularized logistic regression on "a9a".

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Future Direction of C2EDEN

C2EDEN takes small but important step for designing beyond first-order distributed optimization methods. We hope this work can motivate the future work such as:

- Can we design communication and computation efficient second-order methods for **decentralized** optimization?
- Can we design communication and computation efficient higher-order methods for **minimax optimization** or **bilevel optimization**?
- Can we take the advantage of the structure of the Neural Network and the framework of C2EDEN and design efficient distributed optimization methods for training large models?

Q & A

Reference

- Communication Efficient Distributed Newton Method with Fast Convergence Rates. In submission.